

Summary

This project demonstrated a complete basic Arduino circuit setup. An Arduino Uno was programmed and connected via USB to a host computer. On the breadboard, a green LED and a red LED were each wired in series with a resistor and connected to separate digital output pins on the Arduino. The circuit successfully powered and controlled both LEDs. Several practical challenges were overcome during the build, including wiring errors, component polarity issues, and software configuration — all of which provided valuable hands-on learning experience.

Conclusion

Despite the various difficulties encountered, the project was completed successfully, with both the green and red LEDs operating as intended under Arduino control. The process of working through each problem — whether it was a loose wire, an incorrectly placed component, or a software setting — greatly deepened the understanding of how hardware and software interact in embedded systems. This project serves as a strong foundation for more complex Arduino projects in the future, such as incorporating sensors, displays, or wireless communication modules. The skills gained in circuit reading, breadboard layout, and Arduino programming are directly transferable to a wide range of electronics and IoT applications.

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